Roll No.: DL.AI.UM.AID.24.1.20

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2025

Third Semester

Artificial Intelligence & Data Science

Artificial Intelligence & Data Science (Medical Engineering)

23AID204 Advanced Data Structures & Algorithm Analysis

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Develop skills for analyzing algorithmic strategies
CO02	Analyse and apply appropriate algorithmic technique for a given problem
CO03	Implementing standard algorithms on arrays, strings, trees and graphs
CO04	Visualize multidimensional geometry of data structure and concurrency.

Answer all questions

1) Define and differentiate between a full binary tree, complete binary tree, perfect binary tree, and balanced binary tree. Provide one example for each type. [10][CO01][BTL 1]
2) Construct a Binary Search Tree from the sequence [40, 20, 50, 10, 30, 60, 25]. Show the inorder, preorder, and postorder traversals. [10][CO01][BTL 3]
3) Write an algorithm/pseudocode to count the number of leaf nodes in a binary tree using recursion/nonrecursion. [4][CO01][BTL 2]
4) Write the pseudocode of deletion of max heap binary tree. [6][CO02][BTL 2]
5) Consider the array [45, 20, 14, 12, 31, 7, 11, 13].
   (a) Build a Max Heap from this array.
   (b) Perform Heap Sort and show intermediate heap states. [10][CO02][BTL 4]
6) Insert the elements [10, 20, 30, 25, 28, 27, 5] into an AVL Tree.
   Show the tree after each insertion and mention which rotation was applied. [10][CO02][BTL 5]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**10,4,10**	1,2,3	**24**
CO02	**6,10,10**	2,4,5	**26**

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